

MD Jahirul Islam

Building for the next person.

Undergraduate applicant | Engineering | Community



Who I am

A burned school left me with a question: what could someone have done sooner?

I chased it through wires, sensors and Arduino, then noticed the same question in classrooms where students lacked a tutor or exam fees. I build things, test them, and keep coming back to what broke.

500+

marginalized students in a free pathshala

106

cadets led in drill

~20

households in a 1-year flood-alert pilot

300+

trees cultivated and protected

30+

debaters led and coached

The question that started everything

Five moments, one habit: ask what broke.

○ Age seven.

My school near Naogaon burned. Somewhere in it, a notebook with a boat half drawn.

○ New town, same questions.

In Habiganj I kept asking whether it could have been stopped. My grandmother's answer: ask another question.

○ Fifth grade.

Two friends, an Arduino kit, a robot meant to see a flame. One version drove into things. Another froze.

○ Class seven.

A science fair, a broken hand, and a team that stopped hiding failure and rebuilt around it.

○ The next year.

A stronger build and the division crown. The real prize was the gap between the first robot and the last.

When something goes wrong, what can I build so the next person has a better chance?

Engineering & innovation

Robotics, mechatronics, and a flood-alert system.

Robotics

Senior Systems Integration & Design Head, HGHS Robotics & Automation Society

- Challenge** Early robots crashed and froze.
- Action** Architected integrated robot systems and coordinated 16+ peers across 10+ events.
- Result** Division champion.
- Learning** Failure is data. Redesign around it.

Flood-alert research

Independent Sustainable Mechatronics Research Project, HGHS

- Challenge** Flooding that arrived without warning.
- Action** Built a low-cost alert system.
- Result** A 1-year pilot with ~20 households, supporting 2 engineers' community risk work.
- Learning** A design is only finished when someone else can rely on it.

500+

marginalized students learning through Amra O Manush (We Are Humans Too) Pathshala

It began with ten students on plastic chairs in a storage room. We share books, tutor, help with exam fees when we can, and keep showing up.

Why it exists

I saw opportunity vanish for quiet reasons: no tutor, unpaid exam fees, sometimes not even a pencil.

What I do

Founder and lead academics instructor. Books, tutoring and exam-fee help, week after week.

What it taught me

Solutions are rarely handed over finished. They are built through attention, failure and revision.

Leadership in action

Four teams, each with a number to show for it.

106

BNCC

Cadet Sergeant, Drill & Parade Commander

Led 106 cadets in advanced drill and field-commanded 2 National Victory Day parades, reinforcing discipline and public safety for 5,000 community members.

30+

Debate

General Secretary & Public Speaking Coach, Habiganj Debating Society

Led and coached 30+ debaters, organized 8+ tournaments, 2 district titles. Best Debater at an inter-school championship.

15

Football

Team Captain & Defensive Midfielder, Ideal College

Captained 15 players, designed 3x/week drills, and drove the squad to 2 titles and 4 consecutive semifinal runs.

25-30

Interact Club

President, Interact Club of Habiganj

Led 25-30 members: healthcare support for low-income families, advocacy against injustice, volunteers mobilized in local crises.

300+

trees, planted and protected

Founder of the Habiganj Climate Resilience & Tree-Planting Initiative. Over two years, we cultivated and safeguarded trees in schools and flood-hit areas, restoring shade and greenery.

Beyond the trees: healthcare support for low-income families and volunteer response in local crises through Interact Club, and free education through Amra O Manush.

Beyond the classroom

On the pitch

Defensive midfielder, Inter College League squad for Dhaka district.

At the podium

Speaking, and coaching others to speak.

On stage

Lead vocalist at 10+ cultural events for 300+ attendees.

Behind the scenes

Lead organizer for 10+ cultural events: rehearsals, logistics, event-day flow.

How I think

One loop, repeated until it works.

Observe

A school in ashes; students without a tutor.

Question

Could it have been stopped?
What could someone have done sooner?

Build

A flame-sensing robot, a flood alert, a pathshala.

Test

Science fairs, competitions, a year-long household pilot.

Fail

A robot that crashed, one that froze, a broken hand.

Improve

Rebuild around the mistakes until it works.

Share

Teach it, organize it, put it in others' hands.

I am still building.

At university I hope to keep asking questions that start with something going wrong, and to learn how to build better answers alongside people who care about the same problems.

MD Jahirul Islam — Questions first. Then wires.